Auditing P* = 0.21 |R_L|

The law survives. The one result that would have overturned it -- x* drifting with depth in 2D -- was my own cut, and the control I registered caught it.

Three attacks. (1) Is the 4.8x scatter across grammars real? (2) Is the 1D/2D gap just a scale effect? (3) Does x* fall with depth in 2D, as it does in 1D in 15/15 combos?

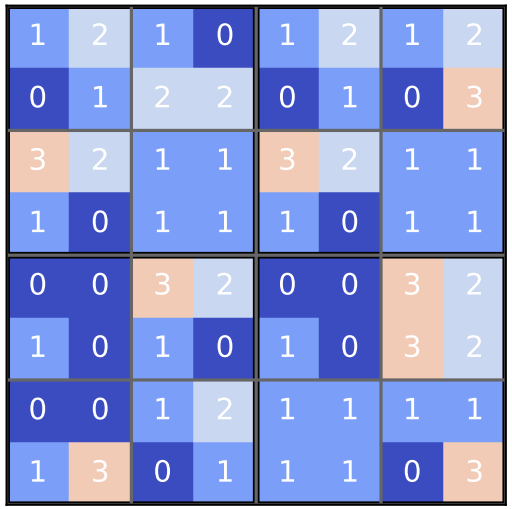
Answers: no, no, and no -- but the third took a new 2D measurement (L=2 m=2 at v=2 and v=3 had never been run) and a defect in how I was cutting the data.

What is new and reusable: the copying regime has a signature computable from every sweep already on disk -- the novelty rate among grammatical outputs, relative to a uniform draw from the support.

Criterion throughout is hshunt's: a grid counts only if it is NOT in the training set AND fully grammatical. Windows are unaligned (block_align=1), so they straddle grammar boundaries.

Tokenization: what a "sample" is, and what the window sees

one sample: 8x8 grid of tokens in {0..3}



Tokens. Every site holds one of v = 4 symbols. There is no pixel intensity and no ordering: token 3 is not "between" 2 and 4. A grid is 8x8 = 64 categorical sites.

Grammar. A root symbol expands 3 times. Each symbol is replaced by one of m = 3 synonymous 2x2 "words", drawn once and then fixed. So the grey lines at left bound aligned 2x2 blocks (one level-3 symbol each) and the black lines bound aligned 4x4 blocks (one level-2 symbol each). A grid is grammatical iff some root derives it.

Diffusion. Absorbing-state ("erasure"): each site is independently replaced by a MASK token with probability p(t). Unmasked sites are never corrupted, so exact Bayes denoising is patch matching -- no score approximation enters.

LS machine. The posterior at a masked site is the empirical distribution of the centre tokens of all training k x k patches that agree with the observed sites inside that window, with Laplace smoothing alpha. Sites are revealed one at a time (exact ancestral sampling; revealing in parallel from marginals adds ~n^2/2T independence errors, which alone made 12% of grids ungrammatical).

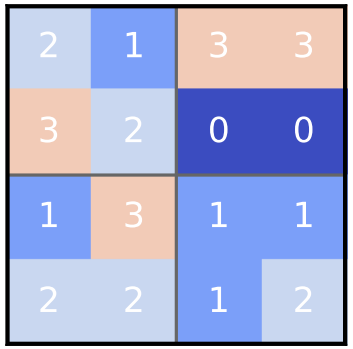
Success (hshunt's criterion). A generated grid counts only if it is NOT in the training set AND parses completely. Since every training grid is grammatical,

$$\text{success} = \text{grammatical} + \text{novel} - 1$$

Example traces: the same machine below, at, and above its peak

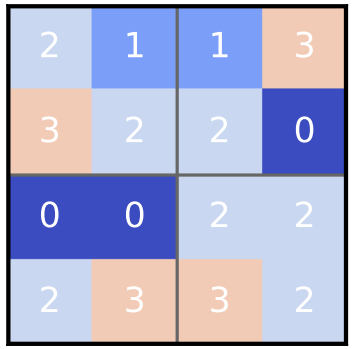
v=4 s=2 L=2 m=2, rule seed 3 -- the cell whose P=3 point outscored its own peak (success 0.0070 against 0.0032 at P=100). At P=3 the machine is grammatical 24% of the time, but 97% of those grammatical draws are one of its three training grids: ratio 0.03 against a uniform draw from the support. That is memorization wearing the label "novel", and an absolute P floor is not the way to cut it -- see the next page but one.

a true RHM sample



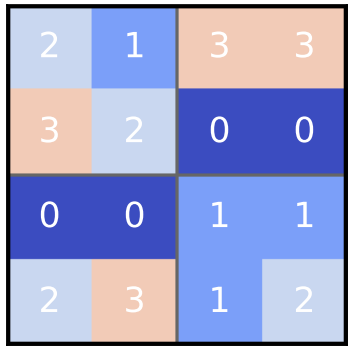
grammatical by construction.
this grammar admits |R_L| = 128
distinct 4x4 grids in all,
so the law puts its peak at
P* = 0.2 |R_L| ~ 26.

LS machine, P=3
(k=3, unaligned windows)



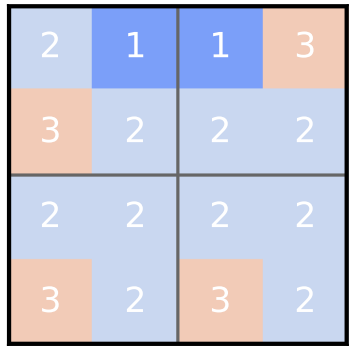
over 16384 draws:
  grammatical 0.2379
  novel 0.7690
  SUCCESS 0.0070
of the grammatical draws:
  novel 0.029
  chance 0.977
  ratio 0.030
most common output:
  11.5% of draws, and it IS
  a training grid

LS machine, P=20
(k=3, unaligned windows)



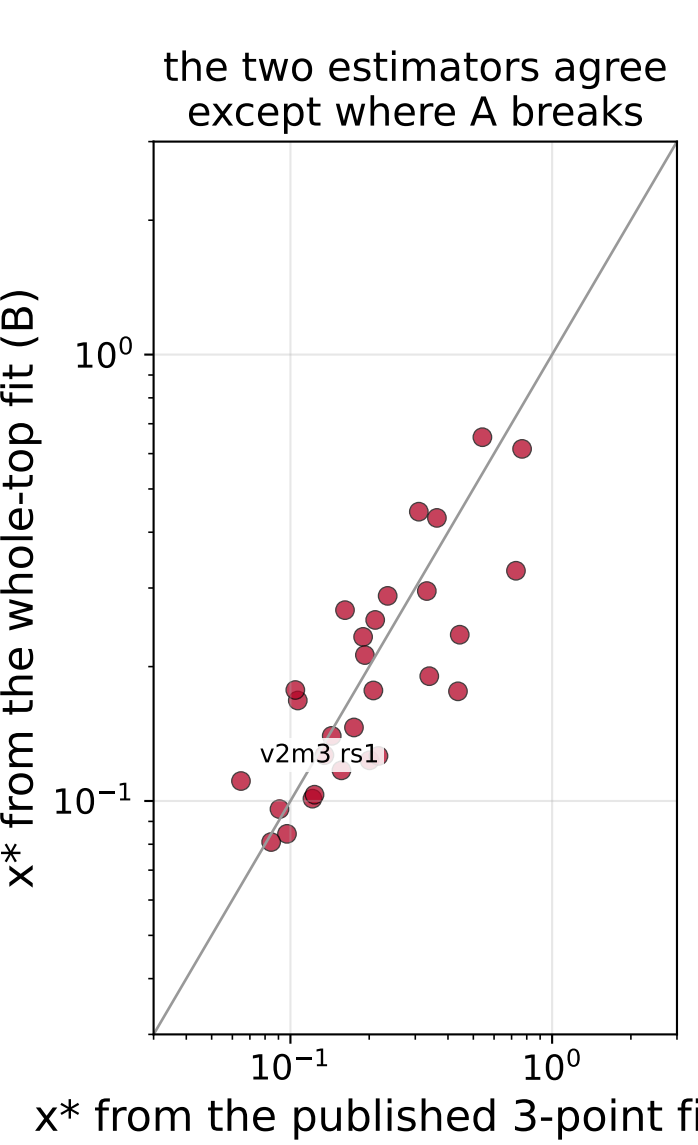
over 16384 draws:
  grammatical 0.0242
  novel 0.9772
  SUCCESS 0.0013
of the grammatical draws:
  novel 0.056
  chance 0.844
  ratio 0.066
most common output:
  0.5% of draws, and it IS
  a training grid

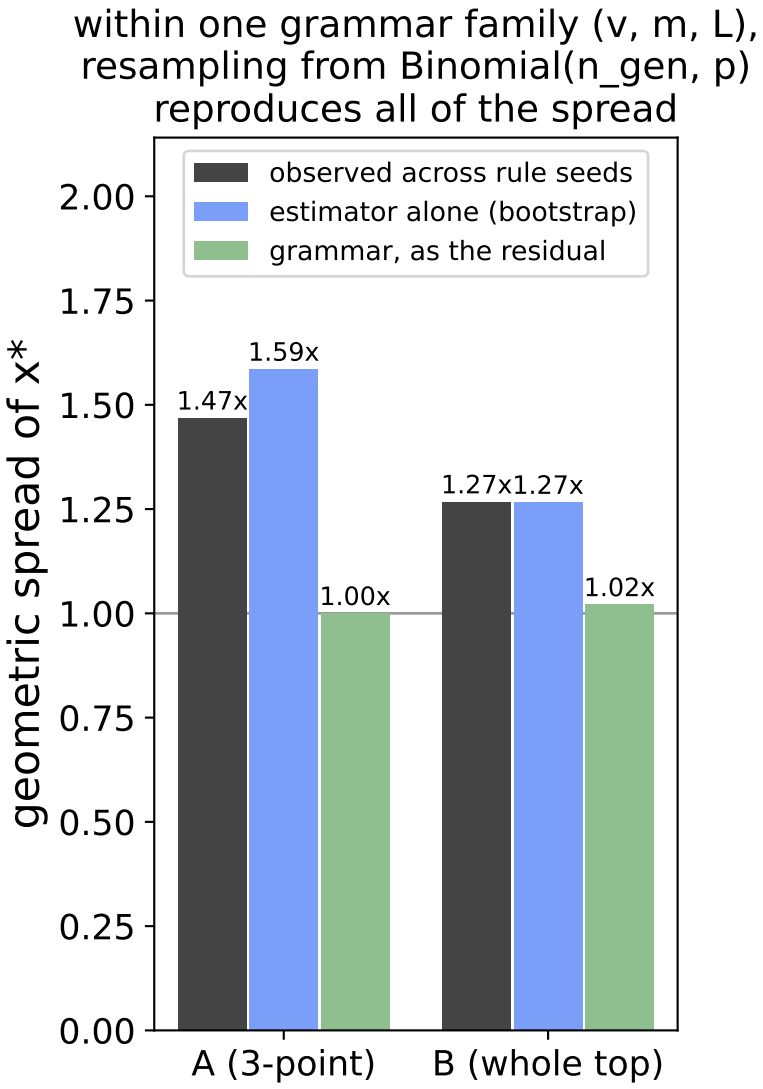
LS machine, P=100
(k=3, unaligned windows)



over 16384 draws:
  grammatical 0.0268
  novel 0.9764
  SUCCESS 0.0032
of the grammatical draws:
  novel 0.121
  chance 0.461
  ratio 0.262
most common output:
  0.1% of draws, and it IS
  a training grid

Attack 1: is the 4.8x cell-to-cell scatter in x* real? No -- it is the peak estimator






```
What A does
  a parabola in log P through the TOP THREE points.
  At n_gen = 4096 a rate near 0.03 has SE ~0.003, so
  the top is a statistical tie and one tied point
  drags the whole three-point window.

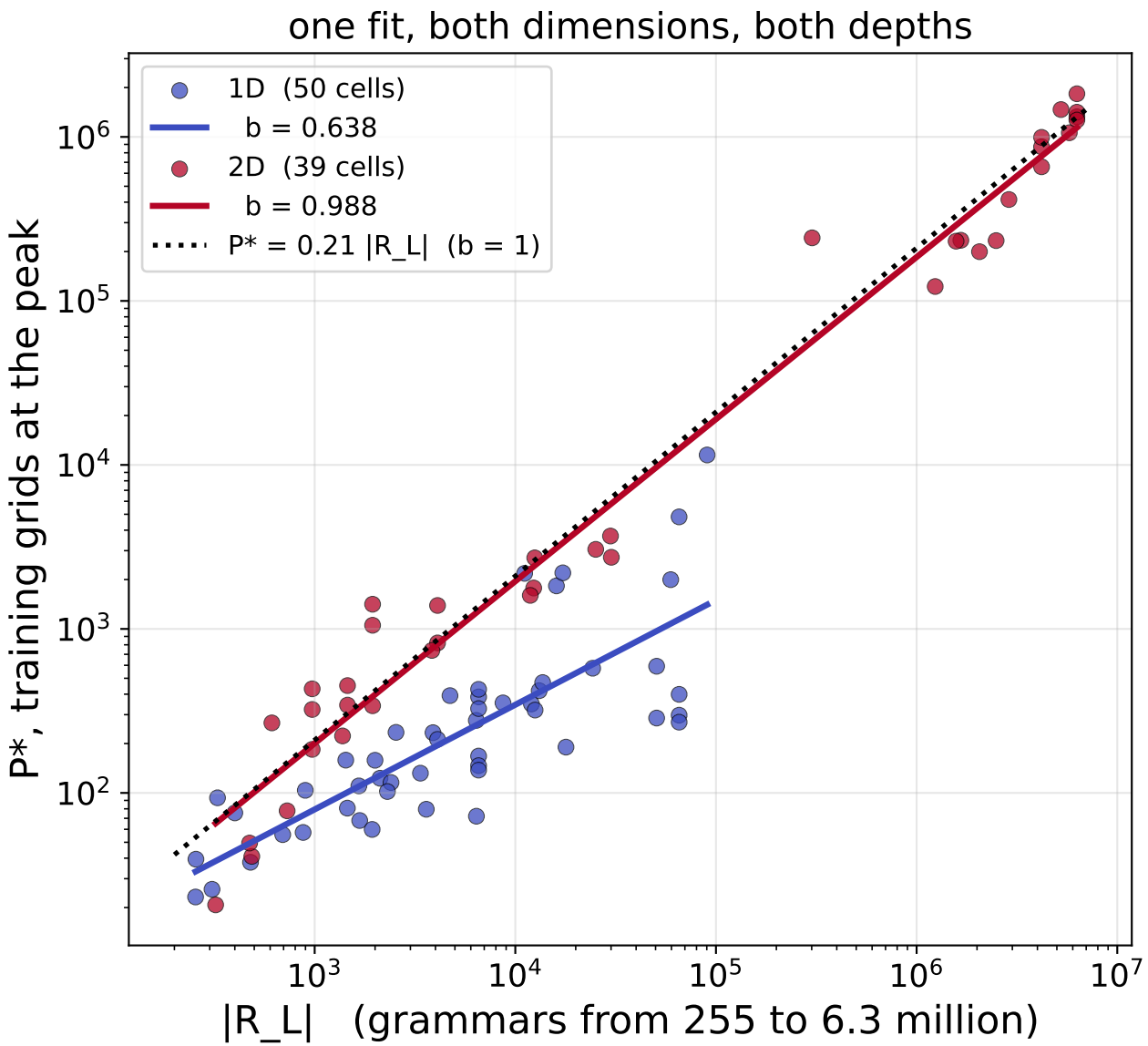
What B does
  inverse-variance parabola over EVERY point within
  2x of the maximum -- 8-12 points on a flat top, so
  no single tie can carry the fit.

The bars group the 30 cells by (v, m, L): same
grammar family, different rule seed, median over the
11 groups.  Log-variances subtract, so the grammar's
own term is the residual -- 1.00x under A, 1.02x
under B.  Undetectable at this power, not zero.

Pooled over all 30 cells the spread is larger (A
1.97x, B 1.77x) because x* moves with (v, m) too.

The headline survives either estimator:
  A   median x* 0.197    b = 0.965 +- 0.041
  B   median x* 0.177    b = 0.927 +- 0.033
```

Attack 2: is the 1D/2D gap a |R_L| scale effect? No -- correcting for it makes the gap bigger



The hypothesis
  x* = |R_L|^(b-1), so b < 1 means "x* falls as
  the grammar grows". 1D cells sit at smaller
  |R_L| than 2D ones -- maybe the 3.87x gap is
  that and nothing more.

The test
  fit log P* = b log|R_L| + c + d*1[2D] over
  every cell, both dimensions, carrying each
  cell's binomial bootstrap through.

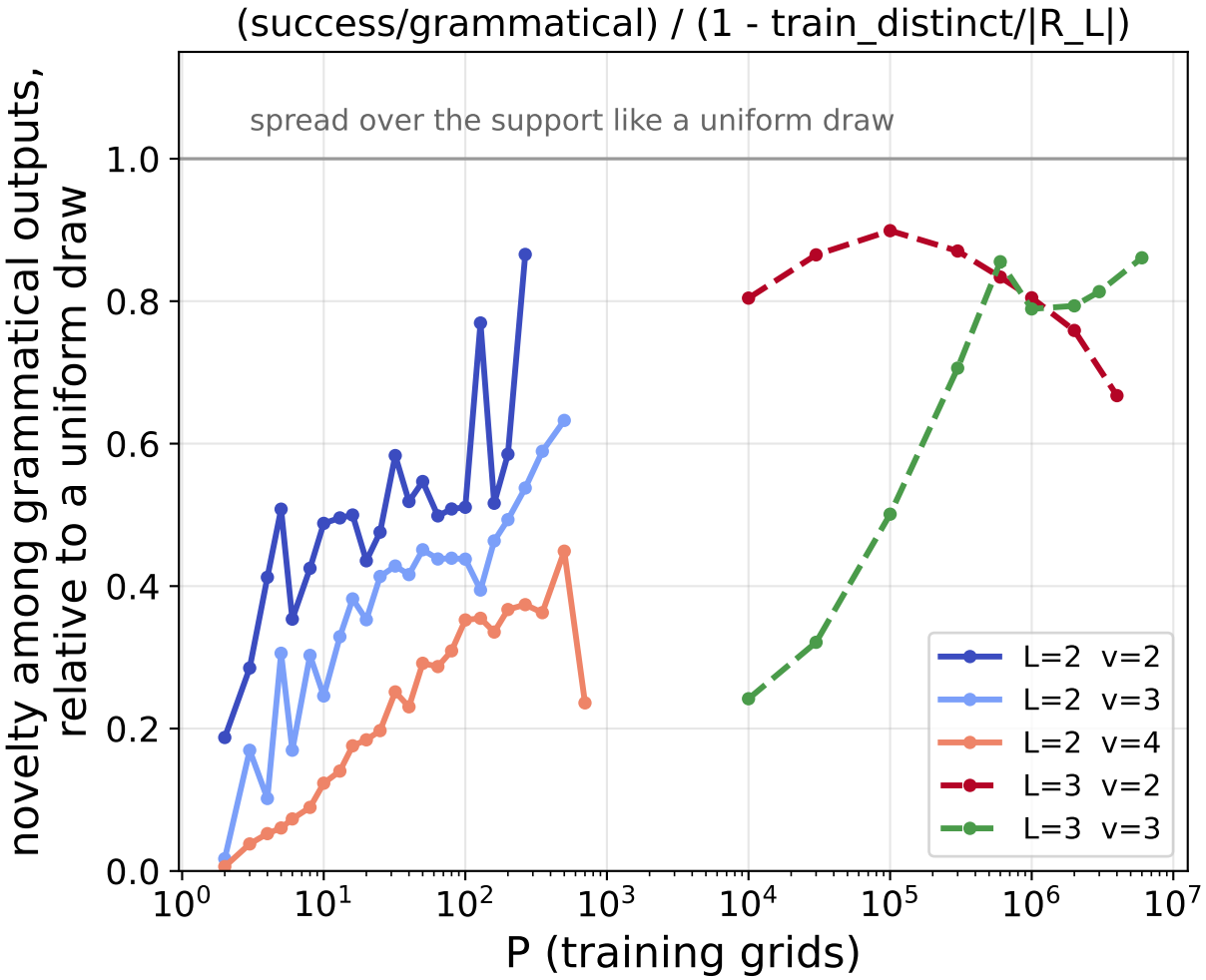
Refuted, and in the awkward direction
  |R_L| >= 200   d = 5.18x in x* [3.44, 8.31]
  |R_L| >= 1000   d = 6.90x
  raw gap   3.87x
  a common b over-corrects the small 1D cells,
  so the offset GROWS to 134-178% of the gap.

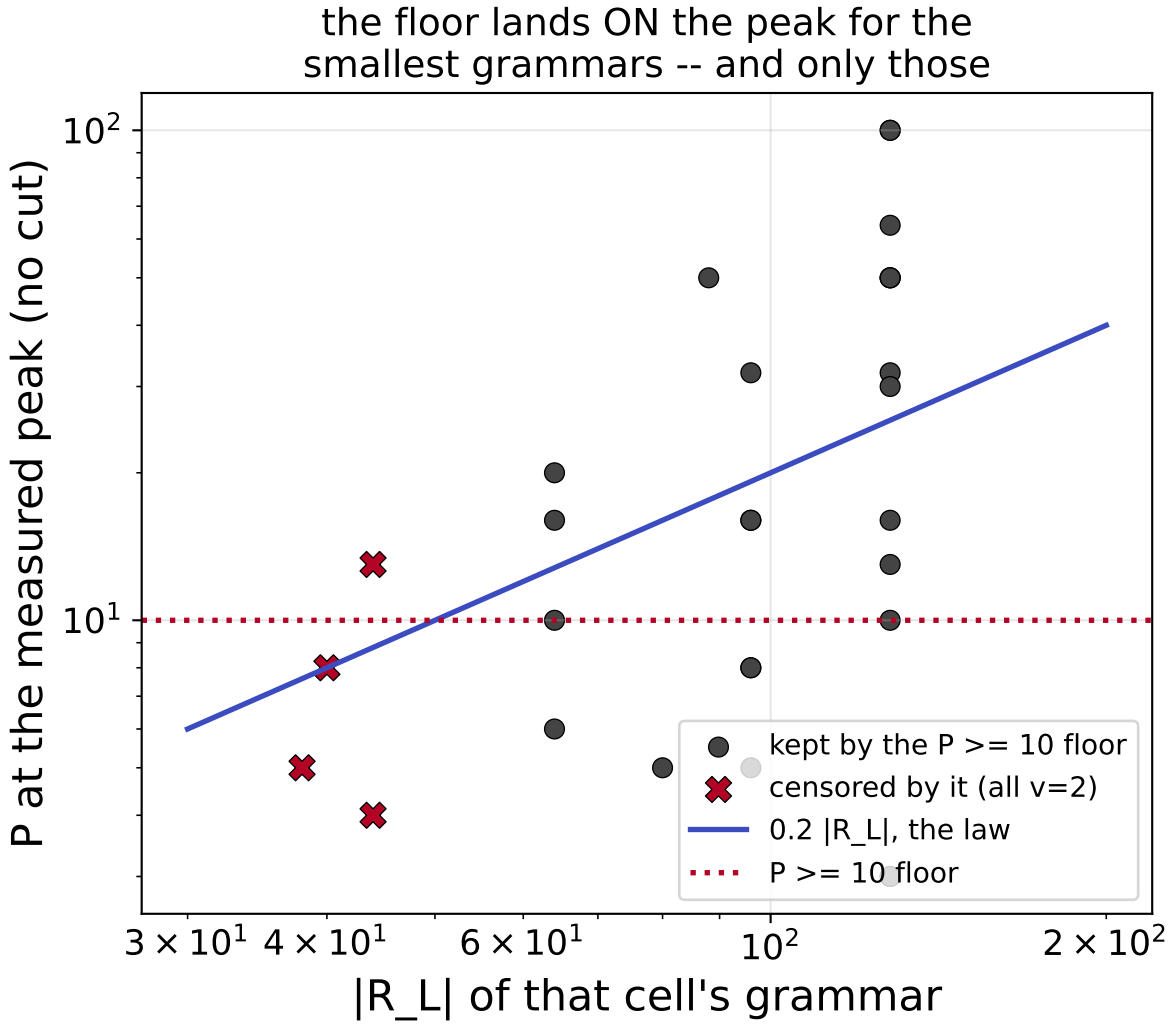
What it does establish
  b_1D = 0.638 vs b_2D = 0.988
  difference 0.350 [0.149, 0.503]
  In 2D x* = |R_L|^-0.012: flat to 1.1x over five
  decades -- THAT is what makes 0.21 |R_L| a law
  in 2D. In 1D the same fit gives |R_L|^-0.36.

Caveat that cost a day
  without the |R_L| >= 200 cut b_1D reads 0.510:
  the P >= 10 floor censors exactly the small-
  |R_L| 1D cells, and that censoring is itself
  |R_L|-dependent. Same defect as page 5.

The copying regime: a signature that never mentions P

With two or three training grids the machine replays them: grammatical, and "novel" only because the training set is too small to match against. The regime is visible in every row of every sweep already on disk, at no extra compute.
A FIXED threshold does not transfer -- L=2 never reaches 1, L=3 does (~0.92) -- so the cut is relative to each cell's own plateau: drop the leading run below half of it. That replaces an absolute P floor whose severity scales with 1/|R_L| -- which is exactly the variable under study.



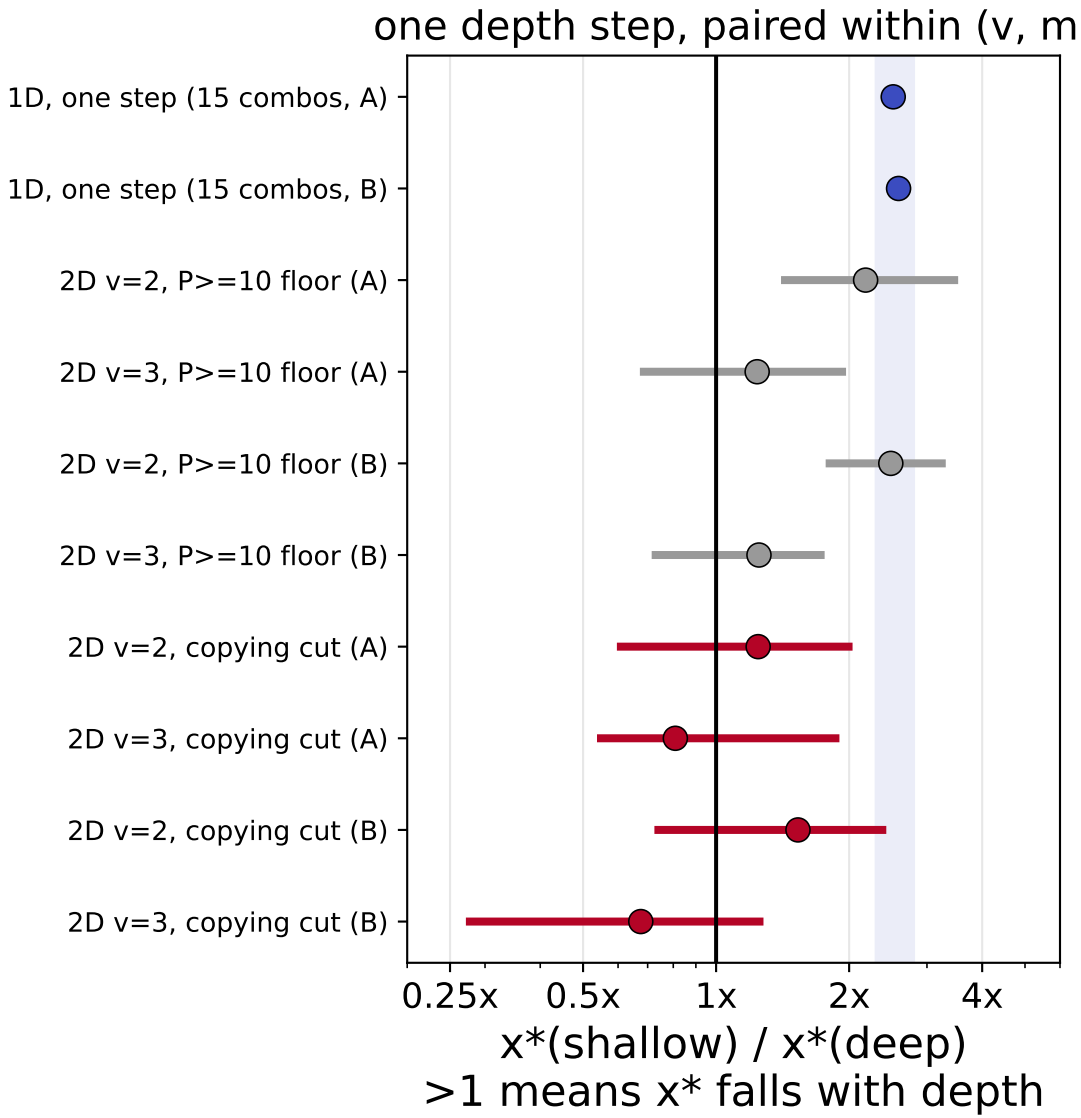


Attack 3: does x* drift with depth in 2D, as it does in 1D? No -- that reading was the floor

Pooling cannot answer this. The published evidence was L=2 pooled over nine (v,m) combos against L=3 pooled over two -- and the two sets are disjoint, while x* varies 7.6x across (v,m). So this pairs within (v, m=2), which needed the 2D L=2 m=2 peaks at v=2,3 to be measured for the first time (job 51244089, 8 rule seeds, n_gen 16384).

Shaded band: the 1D effect. Nothing in 2D reaches it.

The cut has one free parameter. Over copy_rel 0.25 -> 0.8 (both estimators) v=2 moves over 1.21-1.76x and v=3 over 0.53-1.14x, every interval still covering 1.



The bar to clear is 2.5x, not 2.8x
2.8x spans L=2 -> 4, two steps. 2D only has L=2 and L=3, so the matched one-step 1D figure is the comparison: 2.52x (A) / 2.59x (B), falling in 15/15 and 14/15 combos. 1D per-cell estimator noise is 1.09x, so THAT effect is not an artifact.

What the absolute floor did
v=2 m=2 L=2 has |R_L| = 38..64, so P* ~ 8..13. P >= 10 kept 5 of 8 cells and moved their median from 0.190 to 0.311. The three it dropped are the three smallest grammars (P* = 3.9, 4.7, 8.7).

With the copying-regime cut instead
all 8 cells survive, and v=2 falls from 2.18x [1.40, 3.53] to 1.25x [0.60, 2.04] (A), 2.48x -> 1.53x [0.73, 2.43] (B). v=3 runs the other way (0.81x / 0.68x). The v=4 control -- the one cell measured under both old and new settings -- passes either way (1.17x / 0.87x).

And it is now powered, not just inconclusive
51244750 took the L=3 side to 9 seeds at v=2 and 7 at v=3. A 1D-sized step (2.52x / 2.59x) sits OUTSIDE all four intervals, so this is a demonstration that 2D does not do what 1D does, not a failure to resolve one.

And it is what the exponent predicts
b_2D = 0.988 over a 5e4-fold jump in |R_L| gives (5e4)^(1-b) = 1.14x. Two routes, one of them censoring-free, agree.

What the audit changed, and what it did not

UNCHANGED -- the delivered answer

P* = exp(Theta(n)) in the number of sites.
At the U=1 window in 2D, P* = 0.21 |R_L|.
Root needs k >= U1 = 2 s^(L-1) - 1.
The C(k) = f0^U ceiling stays retracted.

It survived a change of peak estimator (0.197 -> 0.177, b 0.965 -> 0.927, overlapping), a change of subsample (b_2D = 0.979 on cells with |R_L| >= 200), and a paired depth test.

SHARPER -- x* is flat, not constant

x* = |R_L|^-0.02..-0.06 across five decades.
Quote 0.21 |R_L| as a law in 2D; in 1D the exponent is -0.36 and there is no law.

STILL OPEN

Why 1D and 2D differ at all. Four mechanisms refuted now: coverage, saturation, entropy budget, and scale. The fourth was the first one measured rather than argued, and it made the gap larger, not smaller.

METHOD -- two defects found, both mine

1. A three-point parabola on a statistical tie. All of the 4.8x cell scatter, and two cells off by 7x and 12x. Fixed: fit the whole top, inverse-variance weighted.

2. An absolute P floor to cut the copying regime. Its severity scales as 1/|R_L| -- the variable under study -- so it manufactured a 2.8x "depth effect" in 2D and dragged b_1D from 0.64 to 0.51. Fixed: cut the regime by its own signature, relative to each cell.

Both defects pointed the same way: toward a more interesting result. That is the warning sign, and it is why the v=4 control existed. It failed first, at 0.42x, and that failure is what exposed the copying regime.

COMPARISON WITH THE PREVIOUS BEST

Same eval, same published population. There is no competing measurement of x* to beat -- this is the first audit of it, and the headline it was auditing is unchanged.

convolutional_diffusion @ 5b40d5a | jobs 51244089 (2D L=2 m=2 at v=2,3, 8 seeds, n_gen 16384) and 51244750 (L=3 seed extension)
scripts/rhm_{peak_noise,copy_regime,depth2d_analysis,xstar_scaling}.py